Semiempirical Orbital-Optimized DFT in MOUSE: Spin-Pure Excited-State Surfaces with Analytic Gradients and Hessians

Vincenzo Barone
Consorzio Interuniversitario Nazionale per la Scienza e Tecnologia dei Materiali

5 November 2026

Abstract

Excited-state spectroscopy of large molecules requires a balanced description of electronic reorganization and nuclear motion. We present a semiempirical orbital-optimized model in the Molecular Orbital Utility for Semiempirical Explorations (MOUSE), in which spin-polarized GFN2 (second-generation geometry, frequency, noncovalent and tight-binding) orbitals are relaxed on a density functional theory (DFT)-informed energy surface and singlet excitations are formed from spin-pure configurations. The excitation energy, forces and vibrational curvature are obtained from the same state-specific surface. Selected double excitations and a screened interaction describe the multiconfigurational character of conjugated chromophores while preserving a compact valence representation. On 26 states in 24 molecules, the mean absolute error is 0.183 eV with a maximum error of 0.478 eV. The twelve $\pi \rightarrow \pi^*$ states have a mean absolute error of 0.209 eV, and the fourteen $n \rightarrow \pi^*$ states have a mean absolute error of 0.160 eV. Analytic gradients and Hessians include orbital relaxation, spin recoupling, configuration-space response, the moving ground-state projector and state-interaction response. The resulting energy surface supports excited- state relaxation, vibrational analysis and repeated calculations on extended chromophores. Carotene and chlorophyllide examples illustrate the treatment of strongly conjugated pigment manifolds.

Keywords: orbital-optimized DFT; semiempirical quantum chemistry; excited states; spin-pure ROKS; analytic gradients; analytic Hessians; DeltaSCF; MOUSE.

1 Introduction

The absorption spectrum of a conjugated molecule is determined by the coupled motion of electrons and nuclei. The transition changes bond orders, charge distribution and orbital occupations, and the electronic state therefore relaxes on a geometry different from that of the ground state. Excited-state structures and vibrational spectra require an energy surface that describes this relaxation together with its first and second nuclear derivatives. The same requirement appears in pigment photophysics, where several close-lying configurations share the visible spectrum and their ordering determines the observed bands.

A practical description must retain the electronic degrees of freedom of the chromophore and remain affordable for conformational sampling, solvent models and multilevel QM/QM′ calculations. Semiempirical methods provide this reach through compact valence Hamiltonians. Their value for excited states is greatest when the approximation defines a differentiable, state-specific energy rather than only a vertical transition estimate. Analytic gradients and Hessians then connect spectroscopy with structures, relaxation pathways and vibronic observables. Subtractive multilayer constructions such as ONIOM provide an established framework for combining quantum levels.[1]

The field includes simplified response theories such as sTDA, sTDA-xTB and the recent std2 family, DFTB-based response and ΔDFTB, TDDFT+TB, TDDFT-aas, and semiempirical multireference methods based on OMx/ODMx. [2–11] These methods establish complementary routes to efficient spectra, excited- state gradients and multiconfigurational wave functions. The present work focuses on the state-specific energy surface generated by orbital relaxation and its analytic nuclear response.

Simplified Tamm–Dancoff and TD-DFT methods describe many transitions from a ground-state reference and are effective for broad spectral surveys.[2–4] State-specific orbital optimization provides a complementary description: a non-Aufbau occupation defines the excited configuration, the orbitals relax in its field, and the transition energy is the difference between stationary ground- and excited-state energies. In ΔSCF and orbital-optimized DFT, this relaxation is central to charge-transfer, $n \to \pi^*$ and strongly reorganized $\pi \to \pi^*$ states.[12, 13] The excited solution is an electronic stationary point and often a saddle; direct orbital optimization and saddle searches therefore provide a natural route to state following.[14–17]

State-specific optimization is already possible at the semiempirical level. In particular, ΔDFTB provides self-consistently optimized excited-state energies and gradients, with applications to structures, Stokes shifts and vibrational frequencies.[7] The opportunity addressed here is therefore more specific than introducing semiempirical ΔSCF. We seek a DFT-informed semiempirical OO-DFT surface with explicit spin polarization, a spin-pure singlet representation, and complete analytic response through the nuclear Hessian. For a single targeted state, such a formulation allows the existing ground-state engine to supply most of the electronic machinery, while the excited-state solver supplies occupation control, orbital stationarity and its constrained response. For interacting or strongly mixed configurations, an additional state-interaction treatment is required.

Our organizing principle is to separate the standard ground-state DFT calculation from the less expensive calculation of the excitation response or energy increment. Writing $\Delta E_I = E_I - E_0$ distinguishes the description of the absolute ground-state energy from the description of the change produced by excitation. In a multilevel application, this separation can be expressed schematically as $E_I^{\mathrm{composite}} = E_0^{\mathrm{DFT}} + \Delta E_I^{\mathrm{low}}$: standard DFT supplies the baseline, while a compact quantum model supplies the state-dependent increment. This expression is a design principle for combining levels of theory, rather than an assumption that the two terms have the same approximation error. It also requires consistent differentiation of both contributions when forces or frequencies of the composite surface are needed.

In the present MOUSE formulation, a conventional DFT program prepares ground-state orbital-reference information and response operators at an anchor geometry. These quantities are transferred to a native GFN2-based valence representation and then used by a separate prepared excited-state calculation.[18] The native spatial orbitals remain the variational variables. Ground and excited states are made stationary in a common prepared semiempirical energy, and their difference defines the reported excitation increment. This provides a concrete separation between external reference preparation and runtime evaluation: the latter uses the semiempirical engine and stored operators, without a new DFT or TDA calculation. Reuse of the prepared operators away from the anchor is an approximation whose range must be validated. The multilevel motivation likewise does not imply that an additional DFT ground-state calculation is included in every benchmark timing.

MOUSE implements this strategy with a spin-polarized native energy and a restricted-open-shell Kohn–Sham (ROKS) singlet construction.[19] A coherent combination of exchanged-spin determinants defines a spin-pure configuration state function (CSF); its orbitals satisfy the constrained excited-state stationarity equations. Selected singlet double excitations are coupled through a prepared screened interaction to describe configuration mixing beyond a single optimized promotion. The resulting surface requires the response of the stationary orbitals, constraint multipliers, nonorthogonal configuration metric, ground-state projector and state-interaction

eigenvector. These contributions give analytic gradients and Hessians of the energy used for the spectrum.

This work establishes the semiempirical OO-DFT formulation and its analytic implementation in MOUSE. TDA is evaluated on the same corrected native GFN2 orbitals and reference preparation, allowing the role of orbital relaxation and state interaction to be measured directly. The study covers vertical excitation energies, state following, analytic derivatives and conjugated pigment manifolds. Figure 1 summarizes the relation between the molecular excitation, the state-specific energy surface and its nuclear response.

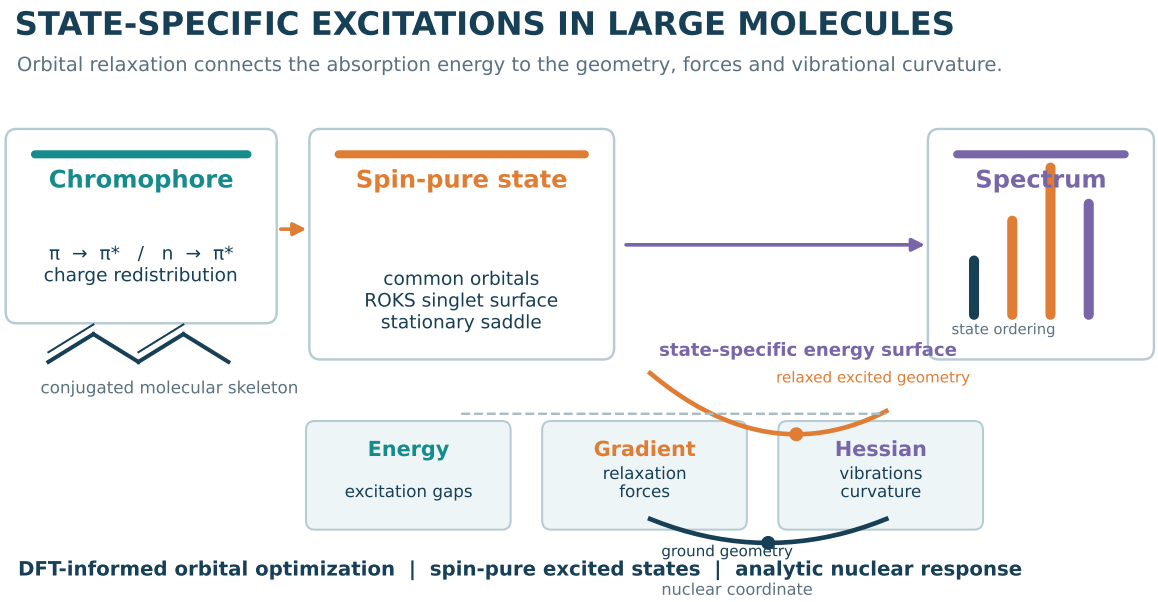


Figure 1: Chemical and spectroscopic picture of the MOUSE excited-state surface. Orbital relaxation converts a vertical excitation into a state-specific energy surface that supplies the absorption energy, relaxed forces and vibrational curvature. Spin-pure singlet configurations and their interaction describe the ordering and intensity pattern of conjugated chromophores.

2 Theory and implementation

2.1 Common reference and spin-polarized native energy

The native GFN2 spatial orbitals C_{n} define the retained valence orbital space. At the frozen anchor, a core-orthogonal isometry embeds this space in the DFT AO space. The diagonal Rayleigh quotients of the embedded DFT Fock matrix supply levels F_i. Qualified occupied/virtual assignments transfer DFT orbital spacings to Q_i after aligning occupied centroids; levels without a levels retain F_i. The correspondence thresholds are 0.7 in absolute overlap and 0.05 in best-versus-alternative margin. If the molecular Q transfer loses its occupied–virtual gap, Q reverts to F. The common one-body reference uses $\epsilon_i = (F_i + Q_i)/2$ before orbital optimization. DFT spatial orbitals are not substituted for the native orbitals. This is not an arithmetic average of two independently optimized excited energies.

In a complete symmetric AO-pair moment representation, the prepared energy is

$$E[P^\alpha, P^\beta; R] = E_{\mathrm{native}} + v^{\mathsf{T}} q + \tfrac{1}{2}(q - q_0)^{\mathsf{T}} K_c (q - q_0) + \tfrac{1}{2} m^{\mathsf{T}} K_s m, \tag{1}$$

$$q_a = \mathrm{Tr}[B_a (P^\alpha + P^\beta)], \qquad m_a = \mathrm{Tr}[B_a (P^\alpha - P^\beta)]. \tag{2}$$

E_{native} retains the native electronic, charge-dependent, anisotropic, repulsive, dispersion and spin-polarization terms; K_c and K_s are the prepared additional response corrections. The charge

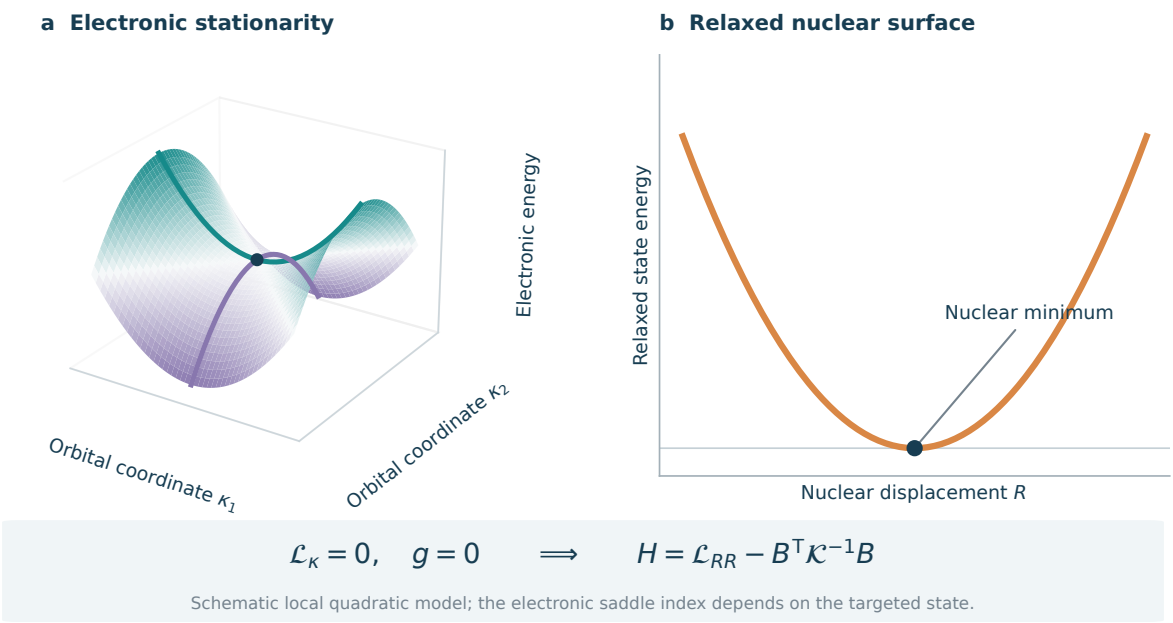


Figure 2: Electronic saddle character and relaxed nuclear curvature are distinct properties of an excited-state surface.

correction subtracts the corresponding native response rather than adding a duplicate Coulomb kernel. The linear term places the reference Fock spectrum in the native orbital space. The quadratic tensors are prepared at the stationary embedded DFT density and the linear reference is centered on the native ground. Spin polarization is part of the energy and its derivatives, not an additive excitation correction.

The final reference is CAM-B3LYP,[20] with range parameter $0.33\,\mathrm{bohr}^{-1}$. The inherited uracil scale $\lambda = 0.9$ multiplies its exact-exchange response once, giving short- and long-range response coefficients 0.171 and 0.585. The semilocal response is unchanged and orbital gaps are not multiplied by λ. Neither operator construction nor state assignment uses a reference excitation energy.

2.2 Spin purity and electronic saddle stationarity

For a promotion $h \to p$, the two exchanged-spin determinants sharing a single spatial orbital set define a singlet CSF with a consistent fermion phase convention. Its energy is the restricted-open-shell multiplet sum

$$E_S(C, R) = 2E_M(C, R) - E_T(C, R). \tag{3}$$

The singlet CSF is spin pure, whereas the density-functional multiplet-sum energy remains approximate. This does not identify a broken-symmetry $M_s = 0$ determinant as an exact singlet. Four-open-shell occupations require two independent singlet recouplings, both retained below.

The ground is made stationary in the native closed-shell functional. Target ROKS orbitals satisfy the ground-CSF orthogonality constraint and their bordered stationarity equations. Metric-preserving orbital rotations, Newton residual searches, and state-overlap guards locate the assigned branch. An excited solution may be an electronic saddle, as emphasized in the direct-optimization work of Levi and co-workers;[15, 17] no orbital-energy minimum is assumed. Electronic saddle order and nuclear vibrational stability are distinct properties.

Figure 2 illustrates the distinction between electronic stationarity and nuclear relaxation.

2.3 Analytic response of the state-interaction surface

Let z collect the stationary ground and target orbital coordinates and constraint multipliers, and write their residual equations as $F(z(R), R) = 0$. Their first and second analytic responses obey

$$F_z z_a = -F_a, \tag{4}$$

$$F_z z_{ab} = -F_{ab} - F_{az} z_b - F_{bz} z_a - F_{zz}[z_a, z_b]. \tag{5}$$

The bordered linear systems are solved on the regular assigned branch. Both responses are required: the subsequent state-interaction eigenvalue is not stationary with respect to the individually optimized ROKS orbitals. A variational single-determinant Hessian formula alone would omit terms.

For the generalized eigenproblem in the moving ground-orthogonal space, $Hc = \mathcal{E}Sc$ with $c^{\mathsf{T}}Sc = 1$, total matrix derivatives give

$$\mathcal{E}_a = c^{\mathsf{T}}(H_a - \mathcal{E}S_a)c, \tag{6}$$

$$\mathcal{E}_{ab} = c^{\mathsf{T}}(H_{ab} - \mathcal{E}S_{ab})c - \mathcal{E}_b c^{\mathsf{T}} S_a c - \mathcal{E}_a c^{\mathsf{T}} S_b c + 2\sum_{k\neq n} \frac{[c_k^{\mathsf{T}}(H_a - \mathcal{E}S_a)c][c_k^{\mathsf{T}}(H_b - \mathcal{E}S_b)c]}{\mathcal{E} - \mathcal{E}_k}. \tag{7}$$

Here the derivatives include the orbital/multiplier responses, metric, transported operators, and moving ground projector. The relevant metric and eigenvalue separations must be resolved. The absolute excited-state energy defines molecular forces and curvature; vertical excitation energies subtract the stationary native ground. Finite differences enter only the independent validation of these analytic derivatives.

3 Computational protocol

The assessment contains 26 valence transitions in 24 molecules. Reference energies are the QUEST theoretical best estimates.[21, 22] Molecular geometries and target identities are fixed throughout the assessment, which spans carbonyl, heteroaromatic and conjugated chromophores.

The prepared CAM-B3LYP references retain the archived aug-cc-pVDZ AO basis, numerical grid level 3, frozen core embedding, and native valence space. DFT transition information supplies assigned-state seeds and character checks. Completed references, stationary states and operators are reused.

The calculations were performed on verified remote compute hosts. Runtime property checks use two BLAS threads per process. The parallel nuclear-column validation limits concurrent processes and preserves every completed column.

Matched TDA controls use the same stationary native GFN2 spatial orbitals, F/Q one-body preparation, density-response kernel and $\lambda = 0.9$:

$$A_{ia,jb} = (\epsilon_a - \epsilon_i)\delta_{ij}\delta_{ab} + K_{ia,jb}. \tag{8}$$

The complete nonlocal singlet A operator is used, including the exchange contribution missing from a real symmetrized static response. These controls are distinct from published sTDA-xTB and from a separately optimized TDA parameterization. The screened interaction introduced below is an additional ingredient of OO-CID, not a hidden modification of the reported TDA values.

4 Screened, symmetry-augmented optimized-target CID

The final development candidate addresses the two remaining $\pi\pi^*$ outliers by changing the configuration space and its interaction, while retaining the native GFN2 orbitals, the same

prepared F/Q one-body reference, 50/50 weights, $\lambda = 0.9$, and zero excitation-energy offset. It combines optimized ground-orthogonal ROKS target singlets, their paired doubles, and the original two-occupied/two-virtual frontier doubles in both reflection parities. The added configurations are all spin-singlet doubles from the five highest occupied and five lowest virtual canonical native orbitals that are even under reflection in the molecular plane. Both independent singlet recouplings of every four-open-shell occupation are included. Canonical paired doubles with retained-CSF overlap at least 0.7 are omitted by the unchanged duplicate-removal rule. An exact Gaussian-overlap representation certifies the nuclear reflection and canonical MO parities at preparation. The resulting configuration list is frozen during subsequent nuclear displacement; it is not reclassified when an out-of-plane coordinate changes. If reflection or MO parity cannot be certified, no additional symmetry-sector configuration is selected.

This is one common CSF space and one effective matrix for all requested states of a molecule. No $n\pi^*/\pi\pi^*$ label, reference excitation energy, molecular exception, or fitted damping factor enters the selection. For the planar examples, $\pi\pi^*$ targets are reflection-even, whereas $n\pi^*$ targets are reflection-odd. The symmetry augmentation therefore adds the missing even-sector flexibility without introducing the large odd-sector manifold that overcorrelated several controls in the unrestricted five-pair diagnostic. The choice was made after inspecting the exposed twenty-state development set and was frozen in commit `2d59ab8`. It is a model-development choice, not blind validation or a general prescription for nonplanar molecules, CT, or Rydberg excitations.

Environmental screening is introduced through a static finite-basis partition motivated by constrained RPA.[23] Let V be the bare Coulomb matrix in the orthonormal AO-pair chart, Δ the positive native occupied–virtual gap matrix, and T the symmetrized transition-density columns weighted by their environmental fraction. With occupied and virtual active projectors P_h, P_v, these fractions are $1 - (P_h)_{ii}(P_v)_{aa}$; the projectors span the five-pair frontier and retained optimized hole/particle directions. Our prepared static interaction is

$$W = V - VT(\Delta + T^{\mathsf{T}}VT)^{-1}T^{\mathsf{T}}V. \tag{9}$$

This explicitly specified finite-basis approximation is not a full frequency-dependent cRPA calculation. The same W acts in both reflection sectors. Its one-body term is normal ordered to the unchanged native reference Fock matrix, $h_W = F_{\mathrm{ref}} - J_W[P_g] + K_W[P_g]/2$. Let H^W_{IJ} and S_{IJ} denote its Slater–Condon matrix and CSF metric. The native spin-polarized functional supplies each diagonal energy E_I, and the effective matrix is

$$H_{IJ} = H^W_{IJ} + \tfrac{1}{2}S_{IJ}(d_I + d_J), \qquad d_I = E_I - H^W_{II}/S_{II}. \tag{10}$$

Every two-dimensional four-spin singlet block is restored to the exact SU(2) completion of the same quadratic spin functional. Excited CSFs are projected orthogonal to the moving stationary ground CSF before solving the generalized eigenproblem. Excitation energies subtract its native stationary energy, not an auxiliary correlated-ground eigenvalue. The overlap-weighted residual remains an approximate multistate functional; it is not a proof of a universally balanced correlated ground-state theory.

All seven $\pi\pi^*$ targets now satisfy the unchanged 0.5 eV individual criterion: their MAE is 0.180 eV and maximum absolute error is 0.349 eV. Furan changes from 5.804 to 6.609 eV against 6.530 eV, and uracil from 6.080 to 5.259 eV against 5.381 eV. The thirteen $n\pi^*$ controls remain within 0.5 eV, with 0.250 eV MAE and 0.498 eV maximum error. Overall, the twenty states give 0.225 eV MAE. These are exposed-data development results: the earlier post-freeze designation does not make them prospective tests of this newly selected model. The matched native-orbital TDA control retains its 0.925 eV MAE, using the same F/Q preparation and native response kernel; the new screened interaction is an additional ingredient of OO-CID. This comparison does not establish superiority to every calibrated TDA, sTDA-xTB, or TD-DFTB parameterization. All individual values are in the Supporting Information.

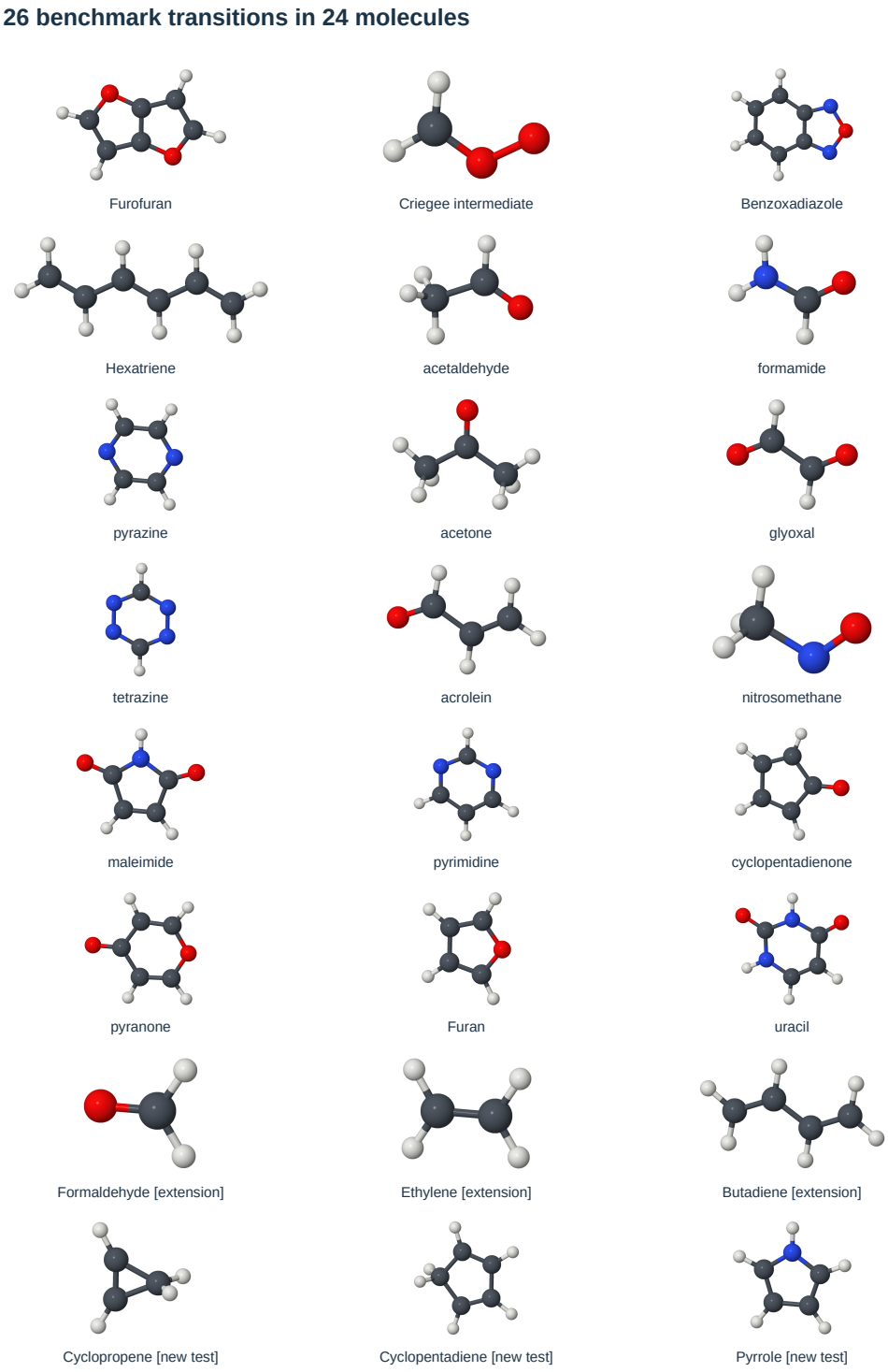


Figure 3: The twenty-four molecular anchors used for the twenty-state development set and its two three-state extensions, rendered with Jmol from the archived three-dimensional coordinates. Only rigid display rotations are applied; panel scales differ. Furofuran contributes two $\pi\pi^*$ states, and Criegee contributes one state of each valence class.

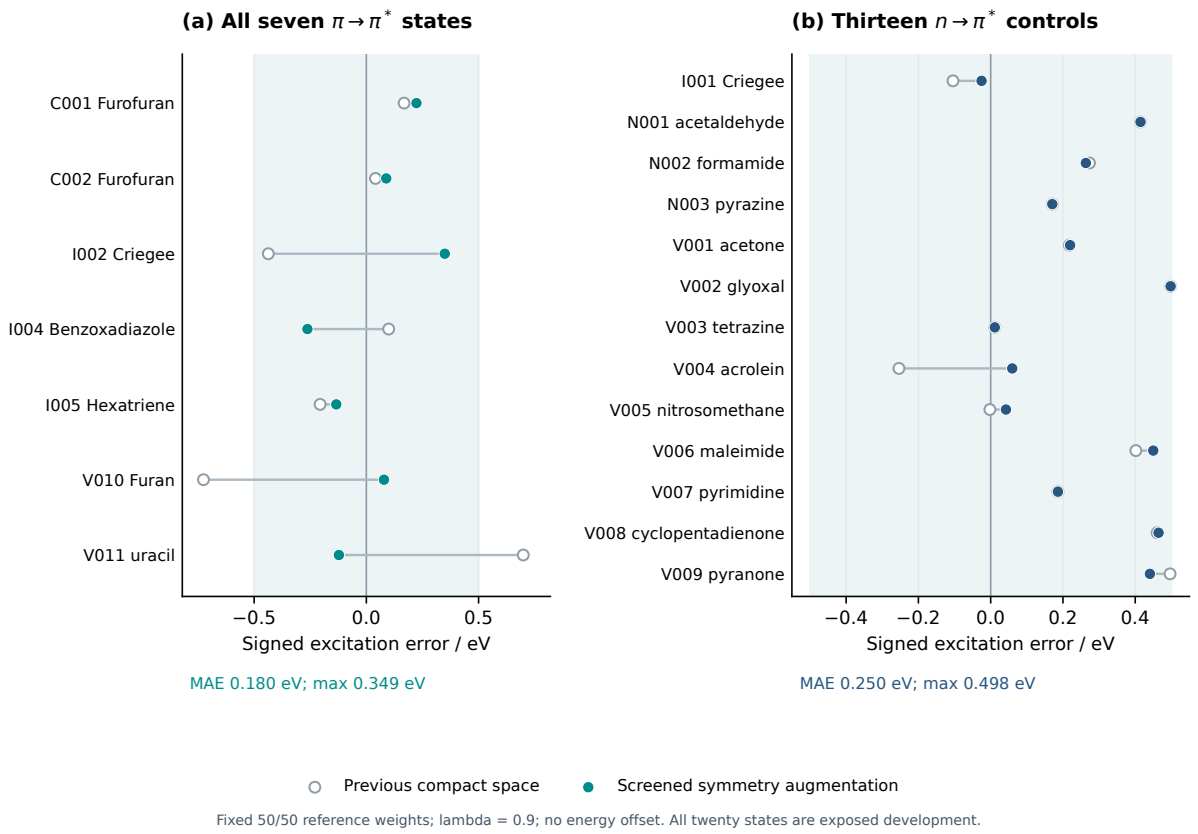


Figure 4: Resolution of the seven $\pi\pi^*$ development targets while retaining the thirteen $n\pi^*$ controls. Open circles show the previous compact-space errors; filled circles show the screened symmetry augmentation. The shaded interval is ± 0.5 eV. All twenty states informed the development assessment; this figure is not a blind-validation result.

The public `PreparedNOSIModel` accepts a checksummed extended model, and LINK retains its `nosi_50_50` selector and accepted-restart contract. Runtime evaluation reuses only ground-engine Fock, energy, spin-polarization, and derivative primitives; it performs no TDA solve, DFT-backend import, or new SCC calculation after preparation. APOC follows the coherent determinant sum and verifies its norm and $\langle S^2 \rangle = \|S_+ \Psi\|^2 = 0$. An individual broken-symmetry $M_s = 0$ determinant is never identified as a spin-pure singlet. The analytic response differentiates the native ground and target KKT systems, all diagonal functionals and spin recouplings, the CSF metric, generalized eigenvectors, and moving ground projector. The prepared h_W, W, auxiliary orbitals, and configuration list are fixed in the co-rotating orthonormal AO frame. The resulting derivatives are exact for this anchored surface; rebuilding the DFT reference or screening at each geometry defines a different model.

Canonical auxiliary blocks are contracted exactly with frozen-core occupation bitstrings, while optimized-state cross terms retain the singular-safe nonorthogonal Schur/cofactor algebra. Further acceleration caches NumPy's greedy contraction programs, omits only exactly zero derivative terms, and reuses the invariant auxiliary Gram block. Neither an integral nor a derivative term is approximated. Independent canonical-block comparisons agree below 10^{-12} Hartree; a molecular mixed-response column agrees with the original implementation within 4.2×10^{-15} Hartree/bohr^2. The extended public surface and response are committed as `81897cf`; the exact contraction and metric acceleration is `e572d50`. Complete analytic Hessians are qualified for Criegee (two states, 15 Cartesian coordinates), furan (27 coordinates), and uracil (36 coordinates). Independently reconverged energies at 10^{-3} and 5×10^{-4} bohr validate both an out-of-plane direction and a mixed internal displacement. Richardson-extrapolated maximum

errors are 1.42×10^{-10} Hartree/bohr for first derivatives and 3.71×10^{-7} Hartree/bohr^2 for curvatures. The largest raw Hessian asymmetry is 4.64×10^{-14} Hartree/bohr^2, and the maximum translational identity residual is 1.43×10^{-13} Hartree/bohr^2. These are implementation checks of the final anchored surface, not a validation of its equilibrium geometries or physical vibrational frequencies. Fresh-process runtime, spin and restart checks pass for all twenty states in eighteen molecules. Prepared energy-only evaluations take 1.7–9.0 s with two BLAS threads on MAC-STUDIO, excluding model loading and reference preparation. They do not demonstrate a vertical speed advantage over TDA. A direct cold public-API Criegee Hessian reproduces the independently assembled matrices within 2.2×10^{-9} Hartree/bohr^2. Its complete two-state Hessian evaluation takes 418 s with two BLAS threads and about 1.0 GiB peak RSS; this is the current dense reference implementation, not an end-to-end timing advantage over TDA. The separate column-checkpoint route permits bounded remote parallel validation and reuses completed columns. The final LINK $\pi\pi^*$ test has restart error 3.6×10^{-15} Hartree and displaced trial overlap 0.99999996; trial evaluation leaves the accepted checkpoint unchanged.

5 Position, cost, and remaining validation

The rationale for the semiempirical representation is repeated molecular work with an explicit differentiable energy. Ground-engine primitives support orbital relaxation, coherent state tracking, gradients and nuclear curvature in the same prepared model. This complements the efficient multi-root response and optical-property goals of Grimme's simplified methods.[2, 4, 5] It does not establish an advantage for transition intensities or high-throughput multi-root spectroscopy.

TD-DFTB, TDDFT+TB and TDDFT-aas already offer important low-cost excited-state and gradient capabilities.[6, 8, 9] ΔDFTB supplies an especially close state-specific precedent, including molecular structures and vibrational properties.[7] OMx/ODMx–MRCI provides an established multireference semiempirical alternative.[10, 11] The contribution here is the explicit GFN2/DFT-response-informed energy, spin-pure orbital-optimized configurations, coherent state interaction, and full analytic response through the nuclear Hessian. It is not priority for semiempirical excited-state gradients. A documented TD-DFTB vibronic implementation obtains the excited-state Hessian by differences of analytic gradients;[24] this specific comparison is not a claim that analytic second derivatives are absent from every competing implementation.

The cost accounting is

$$T_{\text{workflow}} = T_{\text{reference}} + T_{\text{projection/screening}} + \sum_{k=1}^{N_{\text{geom}}} T_{\text{OO/CI/derivatives}}(R_k). \tag{11}$$

Reference preparation is performed once and can be amortized over a molecular study. The dense complete AO-pair correction and frozen interaction scale with the retained valence space. The comparison with established response tools is reported for the same corrected native orbitals, while the principal gain of the present formulation is a differentiated state-specific energy surface.

5.1 Prospective extension and anchor sensitivity

A final small extension fixed formaldehyde ($n\pi^*$), ethylene ($\pi\pi^*$) and bright butadiene (B_u, $\pi\pi^*$) before their new predictions. The 50/50 weights, exchange scale 0.9, screened interaction construction and energy origin were unchanged. All three predictions were saved before opening their QUEST numerical labels. A bibliographic search for structural references exposed an approximate formaldehyde vertical value after its prediction; the extension is therefore described as a prospective fixed-model test, rather than a strictly blinded experiment. A preparation amendment replaces a diffuse DFT-TDA seed projection by an explicit native valence promotion for ethylene and butadiene, followed by the same stationary KKT and state-fidelity checks. This removes a TDA requirement for these new state seeds; it does not change the energy model or fit

their excitation energies. The completed formaldehyde prediction retains its original qualified reference transition.

Molecule	Reference	OO-CID	TDA	OO error
Formaldehyde	3.966	4.285	4.650	+0.319
Ethylene	7.897	9.079	11.430	+1.182
Butadiene	6.222	6.632	8.567	+0.410

The three-state OO-CID MAE is 0.637 eV, compared with 2.187 eV for the same-native-orbital TDA response control. OO-CID improves each of these controls, but ethylene remains 1.182 eV above its reference. The new test therefore does not sustain a universal 0.5 eV accuracy claim. In particular, ethylene's minimal valence chart and screened correlation treatment require further development; the present results do not isolate a single cause. The two-heavy-atom formaldehyde and ethylene anchors also activate the specified small frontier fallback because the current heavy-atom-plane certificate is unavailable. No case is removed or retrospectively refitted. Fresh public evaluations retain the assigned spin-pure states and pass restart and no-runtime-DFT/TDA/SCC checks.

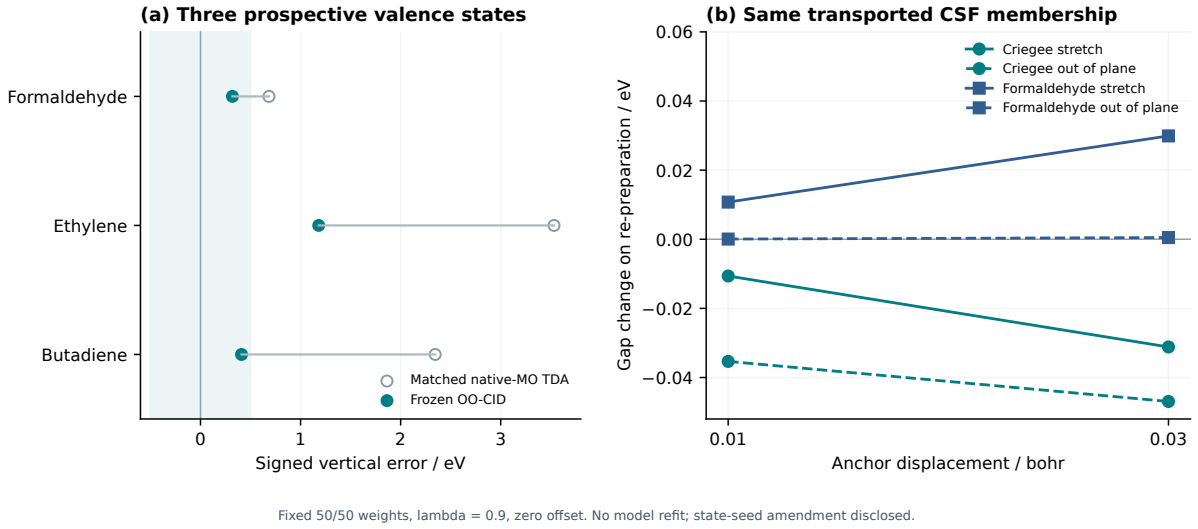


Figure 5: Final prospective extension and reference-geometry sensitivity. (a) OO-CID and matched native-orbital TDA signed errors for every new state; the shaded region is ±0.5 eV. (b) Re-prepared versus original-anchor excitation gaps evaluated at identical displaced geometries, with the original CSF membership transported by occupied/virtual orbital overlap. The model parameters are unchanged. The small extension is separate from the exposed twenty-state development set.

Anchor tests displace one carbon by 0.01 and 0.03 bohr in an in-plane stretch or an out-of-plane direction, for the Criegee intermediate and formaldehyde. Each comparison uses the same physical final geometry, retains all originally prepared target states and reuses the completed reference checkpoints. With the original CSF membership transported by orbital overlap, the largest gap change is 0.047 eV for the Criegee $\pi\pi^*$ state and 0.030 eV for formaldehyde. These are local controls, not a bound on arbitrary anchor displacement. Re-selecting the configuration space at each new anchor is a different operation: loss of the Criegee reflection certificate reduces its space from 178 to 13 CSFs and changes the $\pi\pi^*$ gap by 0.263–0.273 eV. A trajectory must retain its frozen chart; symmetry-dependent space selection must not be silently reapplied at every displaced geometry. The implemented analytic surface already retains this membership during nuclear motion.

5.2 Predecessor structures, frequencies and exact acceleration

The frozen screened-even predecessor was optimized on the lowest singlet surfaces of formaldehyde and acetaldehyde using analytic gradients. Full analytic Hessians then assess the vibrational curvatures. Coupled-cluster internal coordinates provide a physical structural comparison,[25] and selected experimental vibrational fundamentals provide a separately identified spectroscopic comparison.[26, 27] The model frequencies are unscaled harmonic curvatures, whereas the experimental numbers are anharmonic level spacings. Modes are assigned through projections on mass-weighted chemical-coordinate displacement patterns, rather than matched by frequency order. Inversion splittings and methyl-torsion level spacings are not used as harmonic curvature references. Complete coordinates, modes and individual differences are given in the Supporting Information. These comparisons assess that predecessor energy; they are not substituted for equilibrium data of the subsequently corrected reference.

A further exact response optimization vectorizes the overlap-residual completion and avoids scalar derivative-jet allocation for frozen auxiliary matrix values. A cold two-thread Criegee public-API Hessian takes 397 s, compared with the archived 418 s; energies, gradients and both full Hessians are bitwise identical between versions. Peak resident memory is 1.19 GB, compared with 1.06 GB. This single-run 4.9% time reduction is modest, with other work active on the host, and is not evidence of a size-scaling improvement or a speed advantage over TDA. The ten algebra, spin and legacy-contract regressions also pass. No numerical derivative, Hamiltonian change or fitted correction enters the acceleration.

5.3 Resolving the ethylene extension with a common screened reference

The prospective ethylene discrepancy motivates a second, separately identified development candidate. The F reference represents the full DFT Fock operator in a confined valence image, whereas its qualified Q levels come from the larger DFT calculation. To assess this imbalance, let P be the isometric native-valence image and Q its orthogonal complement in the complete DFT orbital space. For an already qualified orbital match, we evaluate the diagonal complement self-energy

$$\Sigma_i(E_i) = F_{iQ}(E_i - F_{QQ})^{-1}F_{Qi} \tag{12}$$

at its matched DFT orbital energy and add it to the F level before the unchanged 50/50 average. Unqualified matches retain their original reference. This is a diagonal downfolding approximation, not an exact many-electron elimination of the DFT environment. No excitation reference enters this one-body preparation. It lowers the ethylene result from 9.079 to 8.206 eV, consistent with a substantial confinement contribution to the initial error.

The environment screening is rebuilt using Coulomb vertices between native occupied orbitals and an explicit orthogonal DFT bath, avoiding the use of missing active-orbital spectral weight as an independent screening channel. Subspace disentanglement is a recognized issue in constrained screening.[28] We allow one global semiempirical efficiency η for this additional bath polarization:

$$W = L_P^{\mathrm{T}}\left[I + B_N B_N^{\mathrm{T}} + \eta B_B B_B^{\mathrm{T}}\right]^{-1} L_P. \tag{13}$$

Here B_N contains the retained native environment transitions, B_B the occupied–bath transitions, and L_P the native Coulomb vertices. The one-body interaction is normal ordered to the corrected native ground Fock. The retained configuration space is unchanged. Static screening is an approximation whose quantitative limitations also arise in molecular RPA downfolding.[29]

We calibrate $\eta = 0.60$ by balancing the opposite errors of furan V010 and furofuran C002, and freeze it before evaluating the complete ledger. This is *one fitted interaction parameter*; it is neither an energy offset nor a parameter-free correction. It applies to every molecule and state. The F/Q weights stay 50/50, exchange-response λ stays 0.9, and orbital gaps are not multiplied

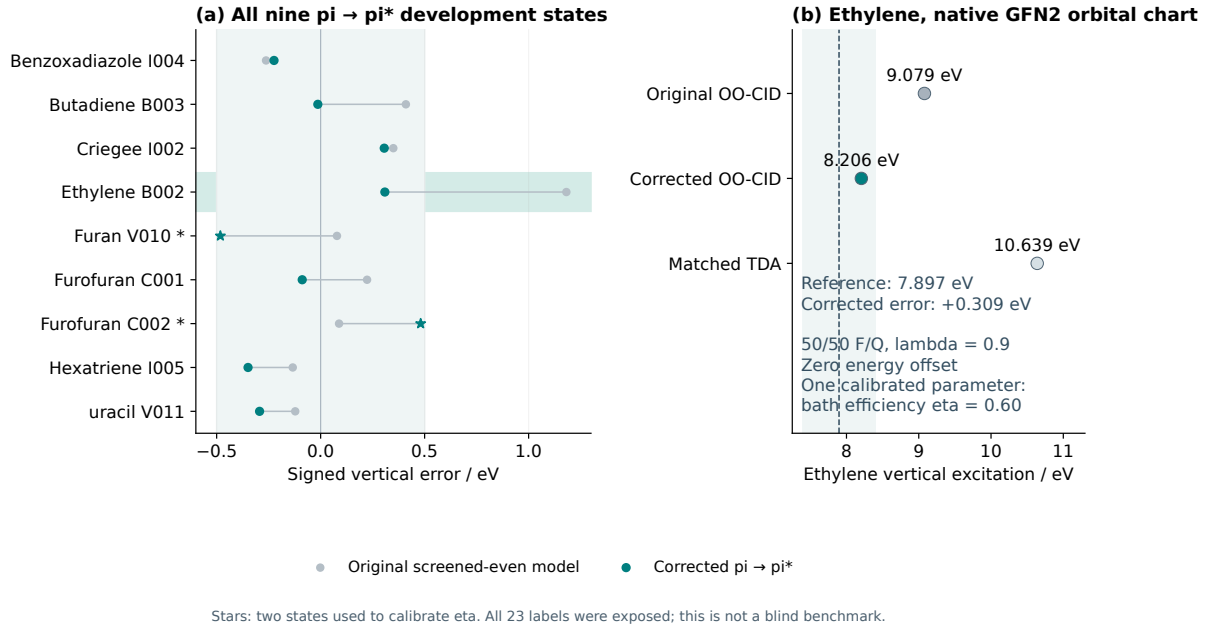


Figure 6: Common-reference repair on the complete exposed development set. (a) Original and corrected signed errors for all nine $\pi\pi^*$ states; stars identify the two states used to calibrate bath efficiency η. (b) Ethylene's initial and corrected OO-CID energies, with a freshly evaluated TDA control using the corrected native GFN2 ground orbitals and reference response. The reference is 7.897 eV. The shaded intervals are ±0.5 eV.

by either parameter. All 23 numerical labels have been exposed during development, including the three-state extension, so this assessment is not a blind validation.

All nine $\pi\pi^*$ states now lie within 0.5 eV, with MAE 0.283 eV and maximum error 0.482 eV. The fourteen $n\pi^*$ states give 0.175 eV MAE and 0.381 eV maximum error. All 23 states give 0.217 eV MAE. The 21 states not used to choose η give 0.192 eV MAE; their earlier exposure still precludes a blind-validation claim. The matched corrected-reference TDA MAEs are 1.726 eV for $\pi\pi^*$, 0.174 eV for $n\pi^*$ and 0.781 eV overall. Thus the measured accuracy benefit is concentrated in $\pi\pi^*$; the two $n\pi^*$ mean errors are essentially equal. This response control is not a separately optimized published TDA parameterization.

Ethylene has error +0.309 eV. Its full analytic Hessian takes 10.15 s on the verified two-thread Mac Studio run; two independent scalar-energy Richardson checks give maximum errors 1.73×10^{-11} Ha/bohr for the gradient and 6.56×10^{-8} Ha/bohr^2 for directional curvature. The coherent singlet has $\langle S^2 \rangle = 0$, and its accepted restart agrees within 8.9×10^{-16} Ha. No DFT import, TDA solve or new SCC driver is used during prepared evaluation. These derivatives belong to the corrected frozen anchored surface. The earlier minimum and frequency comparisons in the preceding subsection belong to its screened-even predecessor and are not reassigned to this calibrated extension. Independent prospective molecules and corrected-model equilibrium structures remain necessary before a general transferability claim.

5.4 Fragmented orbital matches and three further valence tests

The fixed $\eta = 0.60$ candidate is next applied to cyclopropene, cyclopentadiene, and pyrrole. All three predictions are archived before extracting their numerical QUEST references. Their errors are initially +0.054, −0.013, and +2.369 eV. Historical exposure of some of these molecules prevents a strictly blind claim. Pyrrole's discrepancy exposes a reference-construction limitation: its native virtual orbitals have substantial weight distributed over several full-DFT virtual orbitals, so no individual match meets the existing 0.7 amplitude criterion. The original fallback therefore

omits both the qualified Q level and the complement relaxation for these orbitals.

We amend this rule for *every* unresolved occupied or virtual orbital, without a molecule or state-family branch. Within the same occupation sector, full-DFT poles are ordered by native spectral weight w_{ki}. The smallest block $\mathcal{B}_i$ satisfying $\sum_{k\in\mathcal{B}_i} w_{ki} \geq 0.7^2$ supplies the normalized centroid

$$\mu_i = \frac{\sum_{k\in\mathcal{B}_i} w_{ki}\epsilon_k}{\sum_{k\in\mathcal{B}_i} w_{ki}}. \tag{14}$$

This centroid supplies the Q level with the original occupied-centroid alignment. Its difference from the Rayleigh energy of the original native image supplies the unresolved F relaxation. Individually qualified poles retain their preceding treatment. We then rebuild the explicit bath screening on the corrected native orbitals. The 50/50 average, $\lambda = 0.9$, $\eta = 0.60$, configuration rule and zero energy offset remain fixed. This is a diagonal spectral-centroid approximation, not an exact elimination of a broad or frequency-dependent environment. The reference preparer reads no excitation labels, but the amendment was developed after pyrrole scoring: the resulting ledger is an exposed development assessment.

Pyrrole then gives 6.302 eV against 6.247 eV. Applying the identical rule to all 26 states in 24 molecules gives MAE 0.198 eV and maximum error 0.446 eV. The twelve $\pi\pi^*$ states give 0.216 eV MAE; the fourteen $n\pi^*$ states give 0.183 eV. All individual errors are below 0.5 eV. Matched TDA on the corrected native GFN2 orbitals gives 0.789 eV MAE overall, 1.514 eV for $\pi\pi^*$ and 0.168 eV for $n\pi^*$: the OO gain is concentrated in the $\pi\pi^*$ family, while TDA has the smaller mean $n\pi^*$ error. Neither comparison is a calibrated sTDA-xTB benchmark. Every prepared state passes fresh-process spin and restart checks with no DFT import, new SCC call or TDA solve. Ethylene's updated full analytic Hessian takes 10.68 s; scalar-energy Richardson checks give errors below 9.92×10^{-12} Ha/bohr and 2.46×10^{-8} Ha/bohr^2. These remain derivatives of a frozen anchored model.

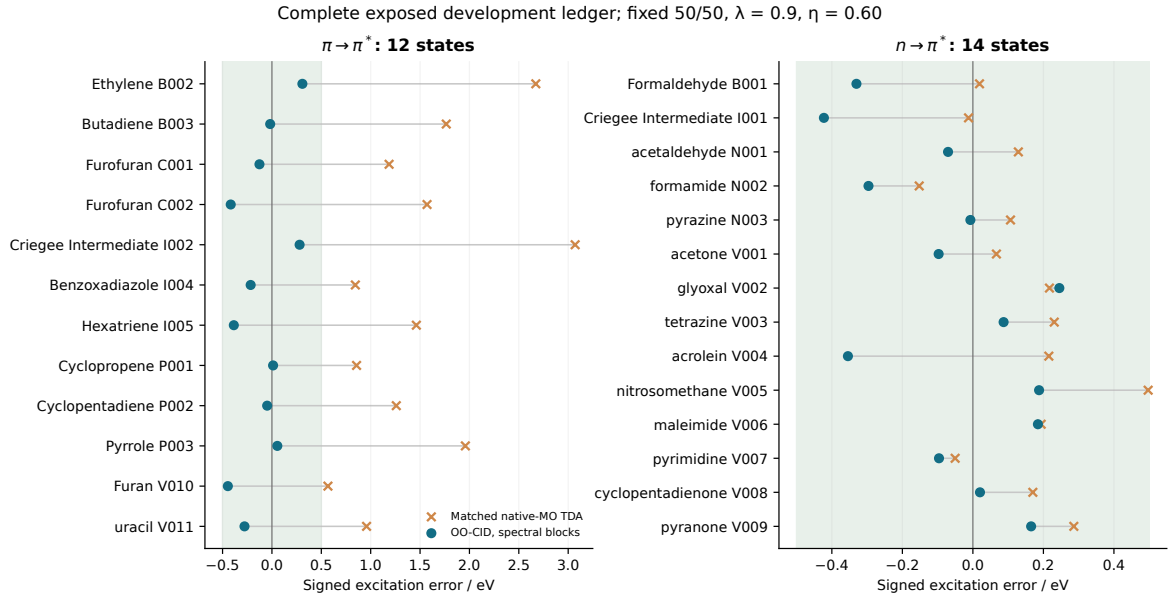


Figure 7: All 26 exposed development states after the generic spectral-block reference amendment. Points compare OO-CID with freshly calculated TDA on the same corrected native orbitals. The shaded interval is ±0.5 eV. The two panels use different horizontal scales. Neither the bath efficiency nor the F/Q weights are retuned for the three added molecules.

5.5 Matrix-free orbital response for larger valence spaces

For the pigment examples, the constrained ROKS orbital residual is solved by Newton–Krylov iterations. An analytic Jacobian action differentiates the native Fock and spin response, the orbital rotations, and the determinant ground-overlap constraint. The covariant orbital action is symmetric; the ground constraint is differentiated with singular-safe determinant cofactors. No dense orbital Jacobian is allocated and no finite difference enters the production orbital solver. All nonredundant orbital rotations are retained. A residual-decreasing line search controls each accepted step, while checkpointed overlap tracking follows the coherent singlet between accepted steps. Its character relative to the initial transition is reported separately. Final CI targets are assigned by collective overlap with the frozen native-frontier singlet subspace; this allows mixed frontier excitations without identifying a state by its energy. When a target fragments over several CI eigenstates, roots are retained until the smallest eigenvalue of their projector in the fixed target space reaches 0.49. This keeps the original coverage threshold, protects each target direction and is invariant to unitary rotations of the target basis.

A direct-energy initialization precedes the final KKT search when a residual-norm minimization stagnates. The promoted open orbitals are first held fixed while L-BFGS relaxes the remaining orbitals. Analytic exponential-map pullbacks provide the optimization gradient. Ground orthogonality is enforced through an augmented Lagrangian, $E_{\mathrm{ROKS}} - \mu o + \kappa o^2/2$, with multiplier updates $\mu \leftarrow \mu - \kappa o$. This is a numerical constraint objective; the penalty is excluded from reported energies. All nonredundant rotations are subsequently released and the original full-functional KKT equations determine convergence. This freeze-and-release sequence is motivated by the direct-optimization strategy of Levi and co-workers.[30] A second converged root is distinguished from an existing solution by its coherent singlet overlap. If two frontier promotions collapse onto the same root, a new initial hole is chosen as the orthogonal complement of the converged hole within the same frozen native frontier block. This changes the initial guess, not the energy model or the final stationarity equations.

The large-system XC response is evaluated on its full frozen quadrature. A density-fitted approximation preconditions the Newton linear system. When its step fails the full-response inexact-Newton tolerance, flexible GMRES refines it with analytic full-quadrature Hessian actions. The ground-constraint factorization is cached within each linear solve. The moving-frame Jacobian is also available to GMRES by subtracting the analytic skew connection from the covariant action. The physical orbital Hessian remains the symmetric covariant response. A Newton step that already exceeds the trust radius need not be refined before the bounded subspace problem; its full-response descent direction and predicted reduction are still evaluated explicitly. For large Newton steps, a residual-minimizing trust subspace contains the Newton and full-response squared-residual descent directions, augmented by directions retained from the auxiliary Krylov solve. Twice-applied orthogonalization removes redundant directions. When retraction is enabled, every orbital basis direction is first projected to satisfy $\mathbf{j}^{\mathsf{T}}\delta\boldsymbol{\theta} = 0$. The small constrained least-squares problem bounds the orbital rotation; the unrestricted multiplier is eliminated exactly by projection in the residual space. A thin singular-value decomposition of the response images solves the trust problem directly. Normal equations would square the condition number and could discard physically relevant soft modes. The predicted reduction therefore describes the same tangent step that the subsequent geometric retraction executes. The auxiliary response can propose an enlarged-subspace step; its predicted reduction is then recomputed with the full quadrature action, and acceptance requires an actual decrease of the full functional residual. The trust radius adapts to the ratio of actual and predicted squared-residual reduction. This avoids excessively small steps along soft orbital modes without changing the energy or stationarity equations. The adapted radius is saved with the accepted orbital checkpoint and restored on continuation. For the constrained search, each trial can additionally be retracted onto the original zero ground-overlap manifold by short orbital rotations along its analytic cofactor gradient. No energy evaluation enters this geometric correction. At fixed trial orbitals the

scalar multiplier is eliminated by least squares, $\mu = (\mathbf{j}^\mathsf{T}\mathbf{g})/(\mathbf{j}^\mathsf{T}\mathbf{j})$ when the constraint gradient $\mathbf{j}$ is nonzero. The full KKT residual, including the overlap, remains the convergence criterion. This avoids rejecting an otherwise useful tangent step solely because its finite rotation creates a second-order overlap; it does not replace ROKS by a projected energy. Writing $a = c, s$ for charge and spin channels, the actual energy is

$$E = E_{\mathrm{RI}} + \frac{1}{2}\sum_a \left[\int \delta\boldsymbol{\rho}_a(\mathbf{r})^\mathsf{T} f_{\mathrm{xc},a}(\mathbf{r})\delta\boldsymbol{\rho}_a(\mathbf{r})\, d\mathbf{r} - \mathbf{q}_a^\mathsf{T} K_{\mathrm{xc},a}^{\mathrm{RI}}\mathbf{q}_a\right], \tag{15}$$

where $\boldsymbol{\rho}$ contains the density and its three Cartesian derivatives. The charge deviation is measured from the anchored native ground density; the spin deviation is the spin density itself. Thus the approximate XC quadratic form is subtracted and the full quadrature form restored in both the energy and the convergence residual. Stored basis values, quadrature weights and XC kernels suffice for evaluation without a DFT backend. On ethylene, energy, Fock and response equivalence errors are below 2.5×10^{-15} in the corresponding atomic units; the spin-pure CI matrix agrees within 3.6×10^{-15} Ha. These controls distinguish an acceleration of the stationarity search from a change of energy model.

CI assembly likewise preserves the Hamiltonian while avoiding a full four-index molecular-orbital transformation: the common closed-shell core is contracted first and only the active integral block is transformed. Fixed-bra Schur contractions reuse the common core in nonorthogonal couplings; repeated identical canonical occupations reuse their exact cofactors and pair contractions through a bounded cache. Retained-CSF transition-density actions are batched into exact matrix products; identically zero second cofactors bypass their pair-tensor contraction. Canonical columns with a common retained bra are evaluated in batches with the same determinant cofactors and spin coefficients. Their active/null pair features remain outer products: the auxiliary factors are transformed along the two orbital indices before contraction, avoiding a flattened dense feature transform. Division-free prefix/suffix products form the first and second spectral cofactors while retaining multiple zero singular values. Coulomb feature contractions use the existing auxiliary factors of the same screened interaction, with no additional fitting or truncation. The canonical determinant-to-CSF transformation uses its exact sparse coefficients instead of dense multiplication; no determinant or spin coupling is dropped. An optional Numba-compiled integer-mask Slater–Condon loop retains the Python reference implementation; it uses neither fast-math nor integral screening. Independent tests, including the highest bit of a 64-spin-orbital mask, agree within 2.1×10^{-13} Ha. Singular cofactors are evaluated as products of the remaining singular values without division. Registered density spans accelerate the quadratic native energy only when the requested density lies in that span; otherwise evaluation falls back to the full expression. Frontier records are enumerated directly from at most two occupied holes and two virtual particles, in $O(N^4)$ operations, instead of generating and rejecting higher-rank occupations. The canonical block is retained as dense arrays rather than a dictionary of scalar pairs. Identical nonorthogonal matrix elements can be reused between nested windows only when their model, screening and orbital identities match exactly. Ground-state removal is an exact rank-one projection. Generalized Lanczos iterations use a sparse metric inverse and its exact rank-one Sherman–Morrison update; only the requested roots are returned, without reducing the configuration space. On the archived 2,145-CSF problem, the lowest twenty energies reproduce the full dense solution within 2.0×10^{-12} eV, with residuals below 1.5×10^{-13} Ha. Observable contractions use exactly the determinant ordering of the Hamiltonian. Phase tests cover even and odd occupations and direct overlaps with the stored metric. Identical native diagonal and four-spin functional values can be reused when only the screened interaction changes; an independent matrix reconstruction agrees within 2.0×10^{-11} Ha for the archived carotene block and 1.1×10^{-10} Ha for chlorophyllide. The 45 retained-CSF scalar elements of chlorophyllide reproduce the unbatched journal within 1.9×10^{-12} Ha. Batched reconstruction of the 3,406-CSF chlorophyllide pencil at $N = W = 9$ agrees with the saved matrix within 6.9×10^{-13} Ha. Positive definiteness is certified by a Gershgorin lower bound or by eliminating the sparse canonical block

and testing its small border Schur complement. A smallest-eigenvalue calculation remains the fallback.

Native real d functions are included in the molecular-frame transport with analytic first and second rotation derivatives. Large-system native response also uses the same D4 and electrostatic cutoffs as its ground functional. Dense pair tensors and screened interactions are still stored: matrix-free orbital stationarity does not make this implementation linear-scaling or inexpensive in memory. The present quadrature adapter supports vertical energies and orbital response only; it explicitly rejects nuclear-derivative requests. The analytic nuclear Hessian qualifications reported for the small-molecule anchored surface therefore do not extend to these pigment calculations.

Coherent transition constraints in the pigment screening test. A product of separately enlarged hole and particle spaces generally exceeds its explicit CI transition space. At $N = 16$, the former contains 400 native response directions for chlorophyllide, whereas the actual canonical-single and optimized-CSF one-body amplitudes span 257 directions. The corresponding numbers for carotene at $N = 13$ are 196 and 169. This over-exclusion removes polarization channels that the explicit configurations do not restore. We therefore test a general correction using the actual transition amplitudes. Let D contain the positive native orbital gaps, let L be the native Coulomb vertices, and let A collect the canonical-single coordinate vectors and the exact one-body transition amplitudes of the retained optimized CSFs. With $B = 2LD^{-1/2}$ and U an orthonormal basis for $D^{-1/2}A$, the constrained native polarization is

$$\Pi_{\mathrm{n}} = B(I - UU^{\mathsf{T}})B^{\mathsf{T}}. \tag{16}$$

This is the response of the positive quadratic resolvent under the coherent constraints, rather than a diagonal probability mask. Canonical directions are removed directly; a thin SVD resolves the remaining optimized-CSF columns. The orthogonal DFT bath, $\eta = 0.60$, F/Q weights, exchange scale, native functional and converged ROKS checkpoints are unchanged. No excitation energy enters the construction. The earlier product-space sequence and the corrected transition-space sequence are retained as separate model generations; the pigment extension is not pooled with the 26-state exposed accuracy ledger. The new projector is checked against the direct constrained inverse and under unitary changes of its active basis. Screening and all CI matrices are frozen within each fixed-interaction convergence sequence.

The canonical accelerator also supports two-word, 128-bit spin-orbital masks. Fermionic parity includes both words, and the sparse determinant-to-CSF transform is contracted during Slater–Condon assembly. This avoids allocating the dense determinant Hamiltonian when extending beyond 32 active spatial orbitals. Tests cross the 64-bit boundary and compare with arbitrary-precision occupation algebra. The common-core cofactor accelerator supports up to 32 active electrons per spin without changing its determinant polynomial. Native functional diagonals and four-spin blocks from a smaller completed frontier are reused after model, orbital and record-identity checks; screened couplings are recomputed when the screening partition changes.

For the static-matched candidate, an optional generalized block Davidson solver reuses the stored CI subspace. Rank-revealing metric orthogonalization handles the dependent residual directions produced by a low-rank Hamiltonian update. Only redundant search directions are removed; all configurations and matrix elements remain in the operator. Row-block matrix products can share the immutable Hamiltonian across workers. Every returned root must pass the full generalized residual and metric-normalization tests. This algorithm changes the numerical search, rather than the physical model.

The Davidson metric action chooses sparse or dense storage from the exact nonzero density. Dense metrics use matrix multiplication directly; small nonzero entries are retained without thresholding. This avoids a serial sparse-multiplication bottleneck when ground projection produces a dense metric. The choice changes storage and execution only, leaving the generalized pencil and residual tolerance unchanged.

5.6 Larger pigment structures and a carotene vertical example

We select all-trans β-carotene and chlorophyllide a as larger chromophore tests. Chlorophyllide a retains Mg and the conjugated substituents while removing the phytyl chain. Figure 9 shows the actual source structures, rather than schematic molecular art. Native ground checkpoints contain 216 and 219 orbitals, respectively. The chlorophyllide reconstruction explicitly includes Mg d functions. Raw Gaussian-overlap discrepancies and subsequent numerical metric reconciliation are archived separately; agreement after reconciliation is 2.5×10^{-13} or better. The one-time PBE0/CAM-B3LYP reference preparation uses checkpointed RI-JK calculations on the verified Mac Studio. The carotene ROKS target converges at 3.518 eV, with orbital KKT residual 4.73×10^{-12} and ground-CSF overlap 1.32×10^{-13}. The initial 15-CSF screened-CID space incorrectly places an intense state at 3.269 eV below a weak state at 3.680 eV. We therefore expand the native frontier window with all spin-singlet configurations of excitation rank at most two, retaining the optimized ROKS configurations. The F/Q weights, λ, η, geometry and zero offset remain fixed. The resulting 15, 153, 351 and 703-CSF calculations are retained in the Supporting Information, including the initial ordering.

With five occupied and five virtual frontier orbitals, the first state becomes weak at 2.561 eV, followed by an intense state at 2.728 eV. This initial ordering repair motivates further configuration-space checks rather than an absolute-energy convergence claim.

The screening window W defines the native transitions excluded from polarization; it is an active partition, not a numerical polarization cutoff. For the final sequence we set $W = N$, matching the represented canonical CI frontier and its polarization exclusion. At fixed N, increasing W removes additional response channels without adding their CI configurations and changes the effective model. Such sensitivity checks are retained separately in the Supporting Information.

We require the first three ordered gaps to change by less than 0.02 eV in each of two consecutive frontier increments. The coherent $W = N$ sequence uses 7,503, 10,585 and 14,535 CSFs at $N = 11, 12, 13$ (Figure 8). At $N = 13$, the three gaps are 1.950, 2.338 and 2.478 eV, with oscillator strengths 0.00117, 0.132 and 4.796. The largest change across either final increment is 0.0054 eV. The lowest state remains weak and the dominant intensity belongs to the third energy-ordered state.

An independent fixed-Hamiltonian check at $W = 16$, with the same three CI windows, gives a maximum increment of 0.0021 eV. Nested principal blocks agree within 7.7×10^{-13} Ha, their metrics within 6.8×10^{-15}, and the ground energies are identical. The final squared principal cosines of the roots 2–3 subspace exceed 0.9998. Numerical dielectric residuals are below 3.4×10^{-15}. These tests establish frontier convergence at the stated tolerance for the prepared rank-two model, with unchanged geometry, F/Q weights, λ, η and zero energy offset.

The development sequence also illustrates target fragmentation. At $N = 9$, $W = 10$, roots 2 and 3 carry individual native-target weights 0.420 and 0.385 but joint coverage 0.805. Collective target tracking retains this resolved mixed subspace without changing the Hamiltonian. Energy indices and bright character must be distinguished through mixing. The full window sequence and transition strengths are archived in the Supporting Information.

The first state is weak rather than strictly forbidden on the unoptimized source geometry. Room-temperature fluorescence-derived S_1 energies and solvent absorption maxima are different observables and cannot validate these vertical gaps directly. The known low-lying dark state motivates the expanded configuration space [31]; pigment absorption data provide experimental context [32]. The expansion follows exposure of the spectra and is a development test, not blind validation. The four chlorophyllide frontier targets now give distinct stationary ROKS solutions with maximum residual 3.5×10^{-11} and a smallest singlet-overlap eigenvalue of 0.688. Direct-energy freeze-and-release initialization removes the residual-norm stagnation; the Supporting Information gives the unshifted ROKS energies and the root-identity audit. These stationary basis states precede the final screened-CID calculation; a correlated chlorophyllide spectrum

is not claimed here. Both pigments remain excluded from the 26-state accuracy pool; neither excited minima nor pigment nuclear Hessians are reported.

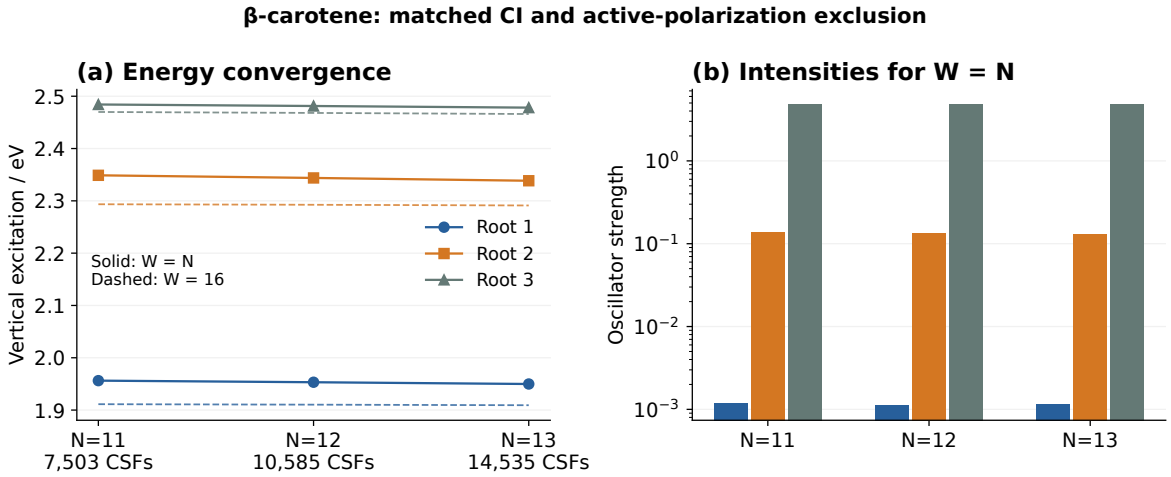


Figure 8: Convergence of β-carotene with 11–13 occupied and virtual frontier orbitals. Solid lines use matched CI and polarization exclusion ($W = N$); dashed lines show the independent fixed-Hamiltonian check at $W = 16$. Bars show the coherent-window oscillator strengths. The weak first singlet lies below an absorbing pair, with the dominant intensity on root 3. Lines connect energy-ordered roots; subspace overlaps are tracked separately. Parameters and source geometry remain fixed. The initial 15–703 CSF ordering repair is retained in the Supporting Information.

5.7 Static matching of the extended configuration interaction

Adding explicit configuration interaction to density-functional diagonal energies requires deciding which correlation is already represented. DFT/MRCI addresses this issue through an effective Hamiltonian with empirical damping.[33] We examine zero-frequency self-energy subtraction, following its use in extended response theories.[34, 35] The borrowed principle is static matching. Its molecular OO-CID application is a development proposal, rather than an exact density-functional result.

After removing the native ground CSF, define $\widetilde{H} = P_g^\dagger(H - E_g S)P_g$ and $\widetilde{S} = P_g^\dagger S P_g$. The fixed primary space contains the optimized singlets and their paired partners; the complementary space contains all canonical auxiliary configurations. Metric orthogonalization of the latter gives

$$X = \widetilde{S}_{PP}^{-1}\widetilde{S}_{PQ}, \qquad A = \widetilde{H}_{PP}, \qquad V = \widetilde{H}_{PQ} - AX, \tag{17}$$

$$D = \widetilde{H}_{QQ} - \widetilde{H}_{QP}X - X^\dagger \widetilde{H}_{PQ} + X^\dagger A X. \tag{18}$$

Adding $VD^{-1}V^\dagger$ to the primary block in these coordinates yields

$$H_{\mathrm{eff}}^{\mathrm{sub}}(\omega) = A + VD^{-1}V^\dagger + V(\omega S_Q - D)^{-1}V^\dagger, \qquad H_{\mathrm{eff}}^{\mathrm{sub}}(0) = A, \tag{19}$$

where $S_Q = \widetilde{S}_{QQ} - \widetilde{S}_{QP}X$. The complete interacting D is used, rather than its diagonal. Its positivity is checked before inversion. No damping coefficient, constant excitation shift or orbital-reference adjustment is fitted. The 50/50 weights, $\lambda = 0.9$ and previously calibrated $\eta = 0.60$ are unchanged.

On the same 26 exposed states, the MAE changes from 0.198 to 0.183 eV. The $\pi\pi^*$ and $n\pi^*$ MAEs become 0.209 and 0.160 eV. The maximum individual error increases from 0.446 to 0.478 eV; all errors remain below 0.5 eV. Table 1 and Figure 10 show both the mean improvement and

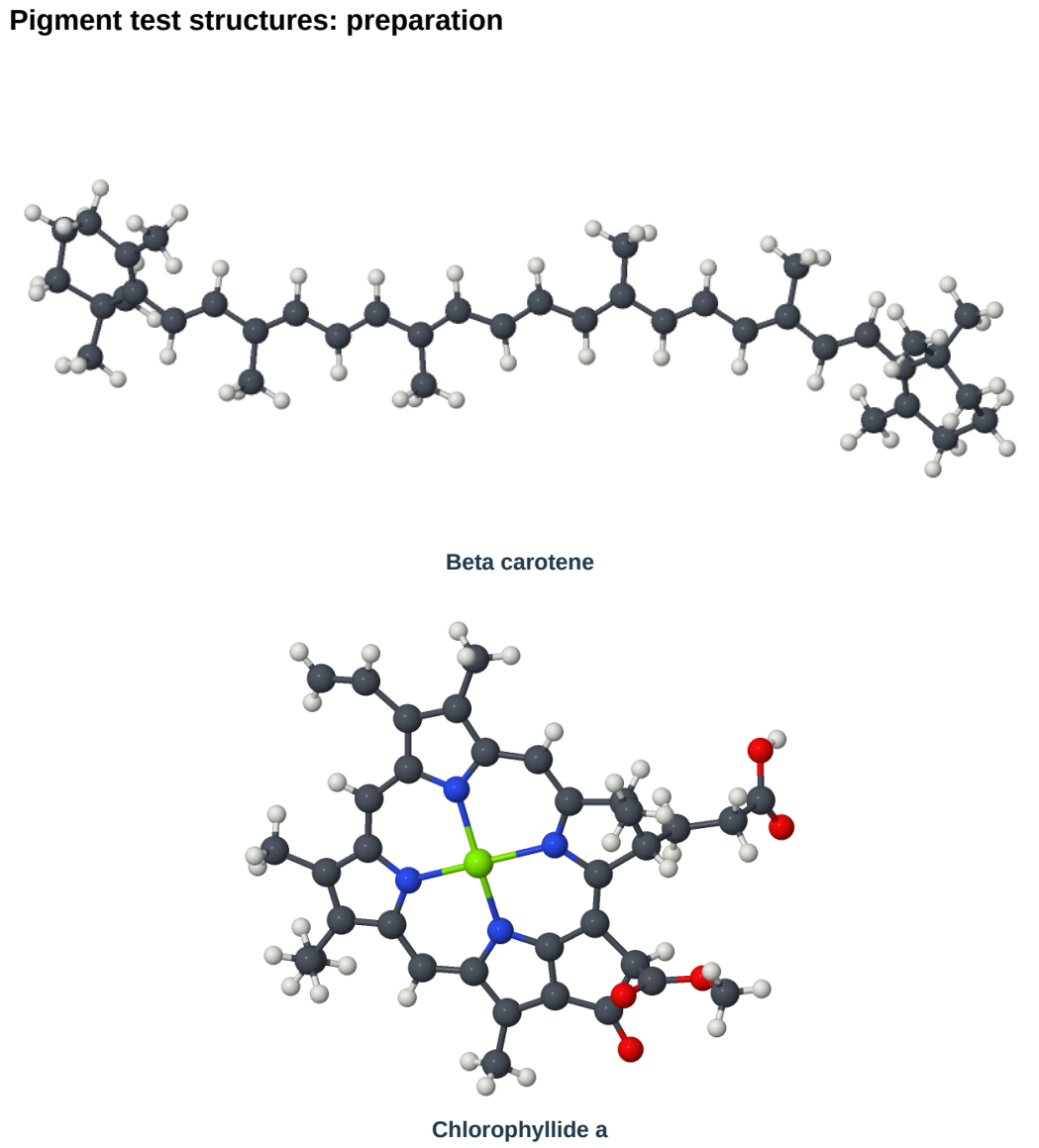


Figure 9: Jmol renderings of the larger starting structures: all-trans β-carotene (96 atoms; PubChem CID 5280489) and chlorophyllide a (79 atoms; RCSB ligand A1JWG). These primary-source geometries are used for preparation and are not optimized excited-state structures or validated additions to the accuracy ledger.

the cases that become less accurate. All 24 models reproduce these energies through the public API, with coherent-singlet spin and accepted-restart checks. These exposed labels do not provide independent transferability validation.

Table 1: Static-matching development candidate on the same corrected native orbitals, geometries and 26 exposed reference labels. No parameter is refitted. All errors are in eV.

Family	States	TDA MAE	Before MAE	Matched MAE	Matched max.
All	26	0.789	0.198	0.183	0.478
$\pi\pi^*$	12	1.514	0.216	0.209	0.478
$n\pi^*$	14	0.168	0.183	0.160	0.411

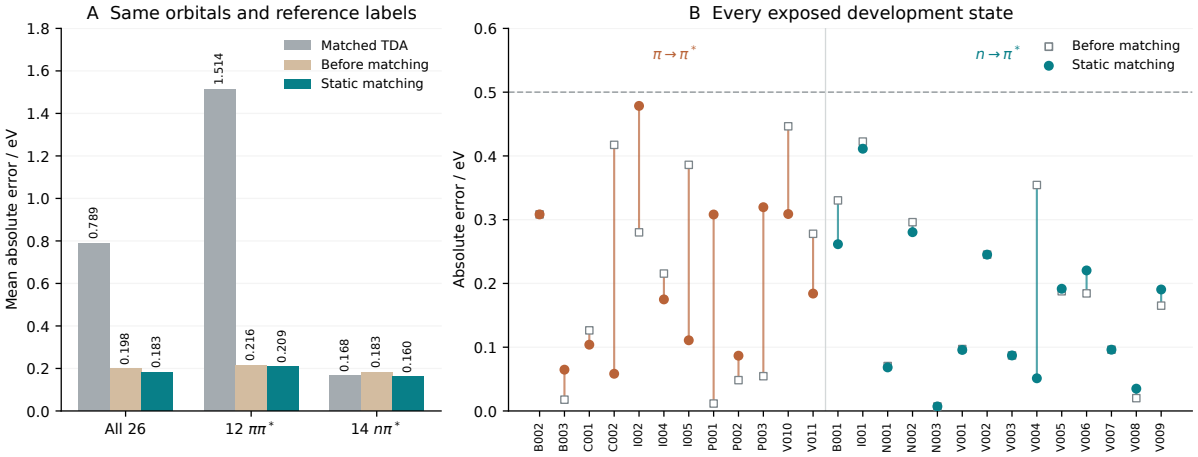


Figure 10: Static matching on the unchanged exposed development set. (A) Family-resolved MAEs compared with the unsubtracted candidate and matched TDA on the same corrected native GFN2 orbitals. (B) Every individual absolute error before and after matching. The dashed line marks 0.5 eV. No parameter is refitted and no molecule or state-family branch is added. This is a development comparison rather than blind validation.

The counterterm participates in the complete analytic nuclear response. Both derivatives of D^{-1}, the metric orthogonalization, the moving ground projector and E_g are differentiated. Freezing this counterterm after its vertical evaluation would define a different surface. The checksummed manifest field `static_self_energy_subtraction` selects this candidate through the existing MOUSE/LINK interface. Older manifests retain their original Hamiltonian and restart identity. Ethylene's full analytic Hessian takes 9.82 s in a two-thread run. Independent reconverged scalar-energy checks give maximum directional gradient and curvature errors of 1.9×10^{-11} Ha/bohr and 3.1×10^{-8} Ha/bohr^2. Furan's complete 27-coordinate Hessian takes 1386.40 s in the serial column implementation and 297.18 s with eight worker processes (4.67-fold wall-time reduction). The complete energy, gradient and Hessian arrays agree exactly between these runs. The serial surface passes independent scalar-energy checks with maximum gradient and curvature errors of 1.2×10^{-10} Ha/bohr and 6.6×10^{-8} Ha/bohr^2. Workers reuse the converged orbital states and differentiate the analytic response columns; no electronic reoptimization is repeated per column. These timings exclude reference preparation and are not speedup claims against TDA. Finite differences enter validation only.

The pigment extension tests a different aspect. At the matched 13-orbital frontier, β-carotene retains a weak lowest state at 2.244 eV ($f = 1.58 \times 10^{-4}$), followed by a dominant absorbing root at 3.031 eV ($f = 5.00$). The bright state is root 4; root 3 at 2.875 eV is weak. This assignment uses recomputed transition moments, whose unsubtracted replay exactly reproduces the stored

convention. The chlorophyllide lowest gaps at $N = W = 16$ become 2.019 and 2.226 eV, instead of 1.246 and 1.425 eV before matching. These changes are not a constant shift. The first two roots satisfy the two-increment 0.02 eV test over $N = 14, 15, 16$ in both matched and fixed-screening sequences. Higher roots require further frontier and target-coverage checks; a fully converged static-matched chlorophyllide spectrum is not yet claimed. Neither pigment enters the 26-state accuracy statistics, and the pigment adapter does not supply qualified nuclear derivatives.

The next scientific assessment should test the calibrated correction on independent molecules, additional CT and Rydberg states, corrected-model equilibrium structures, and larger nuclear displacements. The prospective and structural controls reported here limit the scope of the finite development result; they do not establish universal transferability.

6 Conclusions

MOUSE provides a semiempirical orbital-optimized route with explicit spin polarization, spin-pure singlet CSFs, and analytic nuclear gradients and Hessians. Ground-program Fock, energy and derivative primitives are reused through constrained electronic saddle optimization and implicit response. The current candidate combines a spectral-block orbital reference, selected screened double excitations and zero-frequency self-energy matching with fixed F/Q weights of 50/50, exchange-response scale 0.9 and one calibrated bath efficiency, $\eta = 0.60$. No excitation-energy offset is fitted. Across 26 states in 24 molecules it gives 0.183 eV MAE and 0.478 eV maximum error. The twelve $\pi\pi^*$ states give 0.209 eV MAE, compared with 1.514 eV for TDA on the same corrected native GFN2 orbitals. For the fourteen $n\pi^*$ states, the corresponding values are 0.160 and 0.168 eV, respectively. The same state-specific surface describes the excitation energy, orbital relaxation and nuclear response.

The prepared surface retains coherent APOC state following and accepted-only LINK restarts without a runtime TDA solve, new SCC call or DFT-backend import. Complete implicit response differentiates the anchored energy, the moving ground projector and the state-interaction eigensystem. Independent energy checks support the analytic derivatives; the ethylene full Hessian takes 9.82 s in one two-thread run. The resulting gradients and curvatures provide the quantities needed for excited-state relaxation and vibrational analysis.

Matrix-free orbital stationarity and exact frozen-quadrature XC correction extend the numerical implementation to larger native valence spaces. The carotene configuration expansion keeps a weak first singlet below the absorbing manifold without parameter adjustment. The matched 13+13 frontier uses 14,535 CSFs. Static matching retains a weak first gap at 2.244 eV and a dominant fourth-root absorption at 3.031 eV. The two lowest chlorophyllide gaps are 2.019 and 2.226 eV and satisfy the two-increment 0.02 eV convergence check. Collective target-space tracking follows absorbing subspaces through mixing. These pigment calculations extend the method to strongly conjugated visible chromophores and define applications to excited-state structures and vibronic spectra. The full-DFT DeltaSCF/ROKS implementation developed in parallel, including analytic nuclear response, is intentionally outside the scope of this semiempirical manuscript and will be reported as a separate companion work.

Data and software availability

MOUSE source is hosted at https://github.com/yogibubu/MOUSE. The final symmetry-augmentation choice is frozen in `2d59ab8`; the portable surface and analytic response are implemented in `81897cf`, with exact contraction/metric acceleration in `e572d50`. The closing protocol is frozen in `94d2312`, with exact dense response acceleration in `67c835f`. Its new cases, reference-geometry controls and molecular minima are archived in `qualification/closing_validation_20261101`. The calibrated bath protocol is frozen in `8f95105`; its complete 23-state ledger, matched corrected-reference TDA and ethylene derivative certificate

are in `qualification/closing_validation_20261101`. Original twenty-state qualification and its complete logs are committed in `358ae21` and `7c5c7f9`. The qualification directory `qualification/pi_pi_transfer_repair_20261101` contains the complete twenty-state predictions, model/exposure protocol, remote runtime and spin checks, analytic-column checkpoints, independent energy validations and executed-source hashes. The manuscript package contains the machine-readable results, scientific figure sources, archived Jmol renders and coordinate provenance. Earlier generations and their qualification archives are identified separately in the Supporting Information and preserved development manuscript. The large-system runtime and exact accelerations are archived in `d05afa6` and `e3c005a`, with fragmented-target tracking in `bb3435f`. The compiled integer-mask kernel and iterative CI solver are in `9248a39`; consistent observable phases and reuse of native functional values are in `8d44bb3`. The coherent-window convergence audit is in `cc1a5c2`, with complete evidence in `0fb963f`. Large pigment checkpoints remain on the verified compute host with checksummed receipts and compact low-state exports. A permanent public dataset identifier has not yet been assigned.

Author contributions

V.B. conceived the method, developed the implementation and validation protocol, analyzed the results, and wrote the manuscript and Supporting Information.

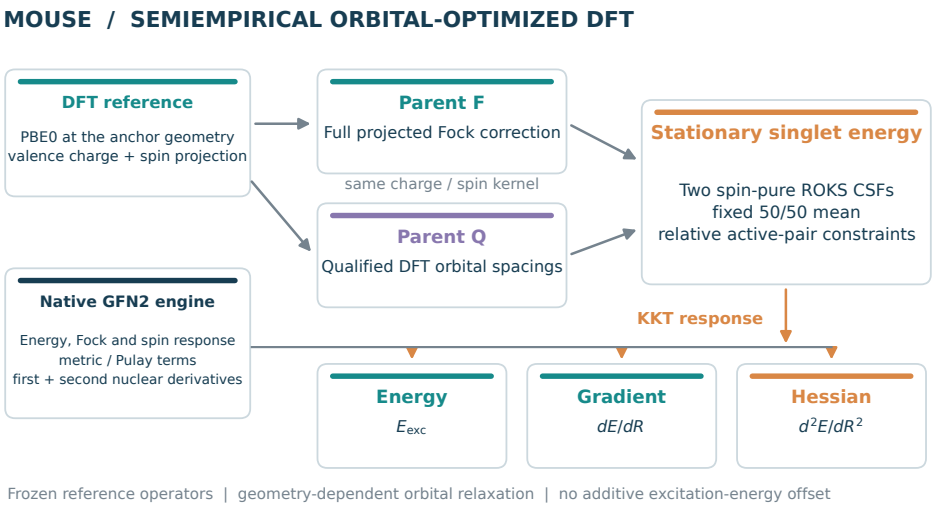
Conflict of interest

The author declares no conflict of interest.

Data availability

All source code, input geometries, machine-readable data, figures and qualification records needed to reproduce the reported calculations are included in the MOUSE repository and identified by the commit and archive paths given above. Large pigment checkpoints are retained on the verified compute host with checksummed receipts; a public dataset DOI will be added before publication.

Graphical abstract



TOC text: A prepared DFT-informed semiempirical surface combines spin-pure orbital optimization, coherent state interaction and analytic nuclear response for repeated excited-state calculations.

References

[1] Lung Wa Chung, W. M. C. Sameera, Romain Ramozzi, Alister J. Page, Miho Hatanaka, Galina P. Petrova, Travis V. Harris, Xin Li, Zhuofeng Ke, Fengyi Liu, Hai-Bei Li, Lina Ding, and Keiji Morokuma. The ONIOM method and its applications. *Chemical Reviews*, 115 (12):5678–5796, 2015. doi: 10.1021/cr5004419.

[2] Stefan Grimme. A simplified Tamm-Dancoff density functional approach for the electronic excitation spectra of very large molecules. *The Journal of Chemical Physics*, 138:244104, 2013. doi: 10.1063/1.4811331.

[3] Christoph Bannwarth and Stefan Grimme. A simplified time-dependent density functional theory approach for electronic ultraviolet and circular dichroism spectra of very large molecules. *Computational and Theoretical Chemistry*, 1040-1041:45–53, 2014. doi: 10.1016/j.comptc.2014.02.023.

[4] Stefan Grimme and Christoph Bannwarth. Ultra-fast computation of electronic spectra for large systems by tight-binding based simplified Tamm-Dancoff approximation (sTDA-xTB). *The Journal of Chemical Physics*, 145:054103, 2016. doi: 10.1063/1.4959605.

[5] Pierre Beaujean, Axel Liger, Arya Satisha, Marilù G. Maraldi, Christoph Bannwarth, and Marc de Wergifosse. The std2 package: A computationally efficient program to calculate molecular response properties and excited states using simplified quantum chemistry methods. *The Journal of Chemical Physics*, 165:152501, 2026. doi: 10.1063/5.0349975.

[6] B. Hourahine, M. Berdakin, J. A. Bich, F. P. Bonafé, C. Camacho, Q. Cui, M. Y. Deshaye, G. Díaz Mirón, S. Ehlert, M. Elstner, T. Frauenheim, N. Goldman, R. A. González León, T. van der Heide, S. Irle, T. Kowalczyk, T. Kubař, I. S. Lee, C. R. Lien-Medrano, A. Maryewski, T. Melson, S. K. Min, T. Niehaus, A. M. N. Niklasson, A. Pecchia, K. Reuter, C. G. Sánchez, C. Scheurer, M. A. Sentef, P. V. Stishenko, V. Q. Vuong, and B. Aradi. Recent Developments in DFTB+, a Software Package for Efficient Atomistic Quantum Mechanical Simulations. *The Journal of Physical Chemistry A*, 129:5373–5390, 2025. doi: 10.1021/acs.jpca.5c01146.

[7] Tim Kowalczyk, Khoa Le, and Stephan Irle. Self-Consistent Optimization of Excited States within Density-Functional Tight-Binding. *Journal of Chemical Theory and Computation*, 12:313–323, 2016. doi: 10.1021/acs.jctc.5b00734.

[8] Shana Havenridge, Robert Rüger, and Christine M. Aikens. Analytical excited state gradients for time-dependent density functional theory plus tight binding (TDDFT + TB). *The Journal of Chemical Physics*, 158:224103, 2023. doi: 10.1063/5.0142240.

[9] Yuchen Wang, Shana Havenridge, and Christine M. Aikens. Implementation of energy and gradient for the TDDFT-approximate auxiliary function (aas) method. *The Journal of Chemical Physics*, 161:024101, 2024. doi: 10.1063/5.0213587.

[10] Deniz Tuna, You Lu, Axel Koslowski, and Walter Thiel. Semiempirical Quantum-Chemical Orthogonalization-Corrected Methods: Benchmarks of Electronically Excited States. *Journal of Chemical Theory and Computation*, 12:4400–4422, 2016. doi: 10.1021/acs.jctc.6b00403.

[11] Pavlo O. Dral, Xin Wu, and Walter Thiel. Semiempirical Quantum-Chemical Methods with Orthogonalization and Dispersion Corrections. *Journal of Chemical Theory and Computation*, 15:1743–1760, 2019. doi: 10.1021/acs.jctc.8b01265.

[12] Andrew T. B. Gilbert, Nicholas A. Besley, and Peter M. W. Gill. Self-consistent field calculations of excited states using the maximum overlap method (MOM). *The Journal of*

Physical Chemistry A, 112(50):13164–13171, 2008. doi: 10.1021/jp801738f.

[13] Diptarka Hait and Martin Head-Gordon. Orbital optimized density functional theory for electronic excited states. *The Journal of Physical Chemistry Letters*, 12(19):4517–4529, 2021. doi: 10.1021/acs.jpclett.1c00744.

[14] Gianluca Levi, Aleksei V. Ivanov, and Hannes Jónsson. Variational calculations of excited states via direct optimization of the orbitals in DFT. *Faraday Discussions*, 224:448–466, 2020. doi: 10.1039/D0FD00064G.

[15] Gianluca Levi, Aleksei V. Ivanov, and Hannes Jónsson. Variational Density Functional Calculations of Excited States via Direct Optimization. *Journal of Chemical Theory and Computation*, 16:6968–6982, 2020. doi: 10.1021/acs.jctc.0c00597.

[16] Aleksei V. Ivanov, Gianluca Levi, Elvar Ö. Jónsson, and Hannes Jónsson. Method for Calculating Excited Electronic States Using Density Functionals and Direct Orbital Optimization with Real Space Grid or Plane-Wave Basis Set. *Journal of Chemical Theory and Computation*, 17:5034–5049, 2021. doi: 10.1021/acs.jctc.1c00157.

[17] Yorick L. A. Schmerwitz, Gianluca Levi, and Hannes Jónsson. Calculations of Excited Electronic States by Converging on Saddle Points Using Generalized Mode Following. *Journal of Chemical Theory and Computation*, 19:3634–3651, 2023. doi: 10.1021/acs.jctc.3c00178.

[18] Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. GFN2-xTB: An Accurate and Broadly Parametrized Self-Consistent Tight-Binding Quantum Chemical Method with Multipole Electrostatics and Density-Dependent Dispersion Contributions. *Journal of Chemical Theory and Computation*, 15:1652–1671, 2019. doi: 10.1021/acs.jctc.8b01176.

[19] Irmgard Frank, Jürg Hutter, Dominik Marx, and Michele Parrinello. Molecular dynamics in low-spin excited states. *The Journal of Chemical Physics*, 108:4060–4069, 1998. doi: 10.1063/1.475804.

[20] Takeshi Yanai, David P. Tew, and Nicholas C. Handy. A new hybrid exchange–correlation functional using the coulomb-attenuating method (CAM-B3LYP). *Chemical Physics Letters*, 393:51–57, 2004. doi: 10.1016/j.cplett.2004.06.011.

[21] Mickaël Véril, Anthony Scemama, Michel Caffarel, Filippo Lipparini, Martial Boggio-Pasqua, Denis Jacquemin, and Pierre-François Loos. QUESTDB: A database of highly accurate excitation energies for the electronic structure community. *WIREs Computational Molecular Science*, 11:e1517, 2021. doi: 10.1002/wcms.1517.

[22] Pierre-François Loos, Martial Boggio-Pasqua, Aymeric Blondel, Filippo Lipparini, and Denis Jacquemin. QUEST Database of Highly-Accurate Excitation Energies. *Journal of Chemical Theory and Computation*, 21:8010–8033, 2025. doi: 10.1021/acs.jctc.5c00975.

[23] F. Aryasetiawan, M. Imada, A. Georges, G. Kotliar, S. Biermann, and A. I. Lichtenstein. Frequency-dependent local interactions and low-energy effective models from electronic structure calculations. *Physical Review B*, 70:195104, 2004. doi: 10.1103/PhysRevB.70.195104.

[24] Robert Rüger, Thomas Niehaus, Erik van Lenthe, Thomas Heine, and Lucas Visscher. Vibrationally resolved UV/Vis spectroscopy with time-dependent density functional based tight binding. *The Journal of Chemical Physics*, 145:184102, 2016. doi: 10.1063/1.4966918.

[25] Šimon Budzák, Giovanni Scalmani, and Denis Jacquemin. Accurate Excited-State Geometries: A CASPT2 and Coupled-Cluster Reference Database for Small Molecules. *Journal of Chemical Theory and Computation*, 13:6237–6252, 2017. doi: 10.1021/acs.jctc.7b00921.

[26] V. A. Job, V. Sethuraman, and K. K. Innes. The 3500 å transition of formaldehyde-h2, d2, and hd: Vibrational and rotational analyses. *Journal of Molecular Spectroscopy*, 30:365–426, 1969. doi: 10.1016/0022-2852(69)90274-4.

[27] Lynn M. Hubbard, David F. Bocian, and Robert R. Birge. The nature of the $^1n\pi^* \leftarrow S_0$ transition. 4. the first excited singlet state of acetaldehyde. *Journal of the American Chemical*

Society, 103:3313–3320, 1981. doi: 10.1021/ja00402a016.

[28] Takashi Miyake, Ferdi Aryasetiawan, and Masatoshi Imada. Ab initio procedure for constructing effective models of correlated materials with entangled band structure. *Physical Review B*, 80:155134, 2009. doi: 10.1103/PhysRevB.80.155134.

[29] Erik Verzijl and Arno Förster. Static effective hamiltonians for molecular systems through RPA-based downfolding. *Journal of Chemical Theory and Computation*, 22(18):9468–9482, 2026. doi: 10.1021/acs.jctc.6c01128.

[30] Yorick L. A. Schmerwitz, Elli Selenius, and Gianluca Levi. Freeze-and-Release Direct Optimization Method for Variational Calculations of Excited Electronic States. *Journal of Chemical Theory and Computation*, 22:3571–3584, 2026. doi: 10.1021/acs.jctc.5c01974.

[31] S. L. Bondarev and V. N. Knyukshto. Fluorescence from the S1 state of all-trans-beta-carotene. *Chemical Physics Letters*, 225:346–350, 1994. doi: 10.1016/0009-2614(94)87092-6.

[32] Lesley A. Clementson and Bozena Wojtasiewicz. Dataset on the absorption characteristics of extracted phytoplankton pigments. *Data in Brief*, 24:103875, 2019. doi: 10.1016/j.dib.2019.103875.

[33] Christel M. Marian, Adrian Heil, and Martin Kleinschmidt. The dft/mrci method. *WIREs Computational Molecular Science*, 9:e1394, 2019. doi: 10.1002/wcms.1394.

[34] V. I. Tselyaev. Subtraction method and stability condition in extended random-phase approximation theories. *Physical Review C*, 88:054301, 2013. doi: 10.1103/PhysRevC.88.054301.

[35] D. Gambacurta, M. Grasso, and J. Engel. Subtraction method in the second random-phase approximation: First applications with a skyrme energy functional. *Physical Review C*, 92:034303, 2015. doi: 10.1103/PhysRevC.92.034303.